

Aprameya Satish

LinkedIn: [linkedin.com/in/aprameya-satish](https://www.linkedin.com/in/aprameya-satish)

Active Secret Clearance

Email : aprameya.satish@gmail.com

Mobile : +1 (470) 263-2882

[aprameya-satish.github.io](https://github.com/aprameya-satish)

EDUCATION

- **Georgia Institute of Technology** Atlanta, GA
Ph.D. in Mechanical Engineering; GPA: 4.0 Aug 2015 – Dec 2020
- **Georgia Institute of Technology** Atlanta, GA
M.S. in Electrical and Computer Engineering; GPA: 4.0 Aug 2017 – Dec 2020
- **BMS College of Engineering** Bangalore, India
Bachelor of Mechanical Engineering Aug 2009 – June 2013

SKILLS

- **Languages:** Python, MATLAB, C/C++
- **Domain Knowledge:** Applied machine learning for acoustics, RADAR, SONAR, and health; statistical signal processing; spatial array processing; computer vision; acoustics and infrasound; elastic wave propagation

EXPERIENCE

- **Georgia Tech Research Institute** Smyrna, GA
Senior Research Engineer Dec 2020 – Present
 - **Vibroacoustic based Pipeline Leak Detection:** Technical lead responsible for experiment design, data management, and statistical DSP and ML algorithm development for pipeline leak detection using vibroacoustic datastreams.
 - **Cognitive Radar Modeling:** Technical lead for ML algorithm development for a new cognitive radar program at GTRI, sponsored by the Missile and Space Intelligence Center (MSIC).
 - **Soldier fitness and health monitoring:** Key contributor in GTRI's signal processing and machine learning work in a multi-million dollar, cross-collaborative effort with Georgia Tech on an ONR-sponsored military health monitoring and prediction project, resulting in multiple publications and a patent.
 - **Infrasound source localization and classification:** Developed a signal processing pipeline that uses measurements from infrasound sensor arrays to localize infrasound targets of interest. Built ML models for infrasound signature classification, and developed a synthetic infrasound signature modeling software suite.
 - **SAR Automatic Target Recognition:** Key contributor in the signal processing and deep-learning based target detection efforts for AFLCMC-sponsored automatic target detection and recognition using SAR data.
 - **Computer-Vision based underwater target detection:** Technical lead on an internal program to build machine learning algorithms for underwater landmark detection from SONAR data for autonomous vehicle localization.
 - **UAV tracking using acoustic beamforming:** Headed algorithm development for outdoor UAV tracking using spherical array acoustic beamforming for a DARPA program.
 - **Infrasound and Acoustic exposure analysis:** Working with WRAIR (US Army), NSWC, and Raytheon on mitigating sound and infrasound exposure to soldiers during weapons and blast training.
 - **Maritime Infrasound Monitoring System:** Coordinated the full wireless communications and signal processing pipeline for a realtime infrasound beamforming dashboard for the ONR.
 - **Internal Research Management:** Secured and managed approximately \$210k of internal funding to develop new technologies in underwater navigation and threat detection using acoustics, and machine learning for health.
- **Georgia Institute of Technology** Atlanta, GA
Graduate Research Assistant Jan 2018 – Jul 2018
 - **Auditory UAV based source localization:** Worked on acoustic source localization in low and very low SNR environments using ground-mounted and Micro-Aerial Vehicle (MAV/drone) mounted microphone arrays.
 - **Monochannel audio source separation:** Implemented a statistical machine learning framework for acoustic source and noise separation based on latent variable decompositions of audio recordings that works with limited training data.
 - **Jet Noise analysis using NMF:** Applied non-negative matrix factorization to study jet engine mode-switching—a combustion instability.
- **SpaceX** Hawthorne, CA
Propulsion Dynamics Intern May 2016 – Aug 2016
 - **Merlin Engine Dynamics Analysis:** Analyzed engine dynamics from pressure, accelerometer, and strain gauge data for the Merlin Stage I and vacuum engines to qualify engine performance. Performed statistical analyses on combustion chamber pressure and peripheral sensor measurements to define nominal operating parameters for Stage 1 and vacuum engines. Automated the process of evaluating engine fatigue using rainflow analyses on strain gauge data.

PHD THESIS

A. Satish, *Analysis and Design of Passive Underwater Acoustic Identification Tags*. PhD thesis, Georgia Institute of Technology, 2020

REFEREED JOURNAL PAPERS

1. D. J. Lin, **Satish, A.**, K. L. Richardson, S. An, C. O. Yaldiz, M. Buller, K. Driver, E. Atkinson, T. Mesite, C. King, O. T. Inan, and A. Medda, "Predicting soldier performance on structured military training marches with wearable accelerometer and physiological data," *IEEE Sensors Journal*, 2023
2. C. O. Yaldiz, M. Buller, K. L. Richardson, S. An, D. J. Lin, **Satish, A.**, K. Driver, E. Atkinson, T. Mesite, C. King, M. Bursey, M. Galer, M. M. Stafford, M. N. Sawka, A. Medda, and O. T. Inan, "Early prediction of impending exertional heat stroke with wearable multimodal sensing and anomaly detection," *IEEE Journal of Biomedical and Health Informatics*, 2023
3. **Satish, Aprameya** and K. G. Sabra, "Passive underwater acoustic identification tags using multi-layered shells," *The Journal of the Acoustical Society of America*, vol. 149, no. 5, pp. 3387–3405, 2021
4. **Satish, Aprameya**, D. Trivett, and K. G. Sabra, "Omnidirectional passive acoustic identification tags for underwater navigation," *The Journal of the Acoustical Society of America*, vol. 147, no. 6, pp. EL517–EL522, 2020
5. **Satish, Aprameya**, B. Nichols, D. Trivett, and K. G. Sabra, "Passive underwater acoustic tags using layered media," *The Journal of the Acoustical Society of America*, vol. 145, no. 1, pp. EL84–EL89, 2019

CONFERENCE PROCEEDINGS

1. S. Das, **Satish, Aprameya**, N. Breen, M. Fitzpatrick, and A. Medda, "Development of synthetic infrasound signatures for target classification using window of opportunity observations," *The Journal of the Acoustical Society of America*, 2025
2. W. N. Newcomb, S. Reddy, **Satish, Aprameya**, and A. Medda, "Time-frequency spectral optimization of underwater sonar acoustic returns for the purpose of characterization using spectral descriptors," *The Journal of the Acoustical Society of America*, vol. 156, no. 4_Supplement, pp. A25–A26, 2024
3. A. Spexet, N. Breen, **Satish, Aprameya**, and D. Alvord, "Intelligent summarization of aviation data for maintenance," in *2024 IEEE Aerospace Conference*, pp. 1–10, IEEE, 2024
4. **Satish, Aprameya**, E. Laubie, J. Casey, and T. Selee, "Automatic target chip extraction from sar images using cnns and cfar," in *Automatic Target Recognition Working Group Meeting*, National Geospatial Intelligence Agency, 2023
5. **Satish, Aprameya** and A. Medda, "Acoustic uav detection using spherical array beamforming," in *2022 Asilomar Conference on Signals, Systems, and Computers*, IEEE, 2022
6. **Satish, Aprameya**, A. Karon, and A. Medda, "Predicting supersonic jet mode-switching using non-negative matrix factorization," *The Journal of the Acoustical Society of America*, vol. 150, no. 4, pp. A177–A177, 2021
7. **Satish, Aprameya**, D. Trivett, and K. G. Sabra, "Passive underwater acoustic tags with curved symmetry for navigation and information encoding," *The Journal of the Acoustical Society of America*, vol. 145, no. 3, pp. 1937–1937, 2019
8. **Satish, Aprameya**, D. Trivett, and K. G. Sabra, "Experimental evaluation of passive underwater acoustic markers," *The Journal of the Acoustical Society of America*, vol. 146, no. 4, pp. 3019–3019, 2019
9. **Satish, Aprameya** and A. Medda, "Time-frequency beamforming adaptation for micro aerial vehicle (mav) noise suppression and source localization," in *2018 52nd Asilomar Conference on Signals, Systems, and Computers*, pp. 884–888, IEEE, 2018
10. **Satish, Aprameya**, B. Nichols, D. Trivett, and K. G. Sabra, "Passive underwater acoustic markers for navigation and information encoding for high frequency sound navigation and ranging (sonar) devices," *The Journal of the Acoustical Society of America*, vol. 142, no. 4, pp. 2731–2731, 2017

PROFESSIONAL SERVICE

- Treasurer, IEEE Atlanta Chapter (*Feb 2022 – Feb 2025*)
- Technical Co-Chair, IEEE SoutheastCon 2024 (*Aug 2022 – present*)
- Reviewer, Measurement (*Dec 2020 – present*)
- Reviewer, Acoustical Society of America (*Dec 2020 – present*)
- Grant Committee Member, Melton Foundation (*Dec 2020 – present*)
- Grad Gala Operations and Logistics Chair, GT Graduate Student Association (*Mar 2020*)
- Treasurer, Acoustical Society of America GT Student Chapter (*Dec 2017*)
- President's Undergraduate Research Award (PURA) Grant Committee Technical Reviewer (*Jun 2016, Jun 2020*)
- Co-Organizer and Compliance Officer, TEDxBangalore (*Aug 2014*)
- Senior Fellow, Melton Foundation (*Jun 2013 – present*)

RECOGNITION

- Outstanding Service Award, IEEE Atlanta Section, (*Oct 2024*)
- Outstanding Service Award, GT Graduate Student Association, (*Mar 2020*)
- GTRI Best Internally Funded project (IRAD) nominee, (*Jun 2018*)
- Mercedes-Benz RD Star Performer nominee, (*Jun 2014*)